

Hadoop Installation and Basic Usage Guide

Prerequisites: Linux (Ubuntu), Java JDK 8+, SSH, 4 GB RAM.

Step 1: Install Java

```
sudo apt update && sudo apt upgrade -y  
sudo apt install openjdk-8-jdk -y  
java -version
```

Step 2: Create Hadoop User

```
sudo adduser hadoop  
sudo usermod -aG sudo hadoop
```

Step 3: Configure SSH

Install SSH, generate SSH keys, and enable password-less login.

Step 4: Download and Extract Hadoop

Download Hadoop from Apache, extract, move to /usr/local/hadoop, and set ownership.

Step 5: Configure Environment Variables

Set JAVA_HOME, HADOOP_HOME, HDFS, YARN, and PATH variables in .bashrc.

Step 6: Configure Hadoop XML Files

Edit core-site.xml, hdfs-site.xml, mapred-site.xml, yarn-site.xml, and hadoop-env.sh.

Step 7: Format NameNode

Create namenode and datanode directories and run: hdfs namenode -format

Step 8: Start Hadoop Services

start-dfs.sh, start-yarn.sh, and verify using jps.

Step 9: Access Web Interfaces

NameNode: <http://localhost:9870>
ResourceManager: <http://localhost:8088>

Basic HDFS Commands

```
hdfs dfs -mkdir /input
hdfs dfs -put sample.txt /input
hdfs dfs -ls /input
hdfs dfs -cat /input/sample.txt
hdfs dfs -get /input/sample.txt
hdfs dfs -rm /input/sample.txt
```

Running WordCount Example

Upload a text file and run the Hadoop MapReduce WordCount example using the examples JAR.

Hadoop Architecture

HDFS stores data. NameNode manages metadata. DataNodes store blocks. YARN manages resources. MapReduce processes data.

Prepared for academic and laboratory use.